Household Animals Classification Using Deep Learning

Mini-Project Report | Image Classification: Cats vs. Dogs

1. Abstract

This project implements and compares two deep-learning approaches for classifying household animal images into cat and dog categories. The first approach is a custom Small Convolutional Neural Network (CNN), trained from scratch. The second uses a VGG-16 network pretrained on ImageNet as a frozen feature extractor, with a newly trained classifier. The recorded validation accuracies were 87.80% for the 20-epoch Small CNN and 98.80% for the VGG-16 feature-extraction approach. The results demonstrate the potential benefit of transfer learning for this image-classification task.

2. Dataset and Preprocessing

The dataset contains 25,000 labeled images: 12,500 cats and 12,500 dogs. The experiments used 20,000 images for training and 5,000 for validation. The Small CNN resizes images to 128 × 128 pixels. The VGG-16 pipeline resizes images to 224 × 224 pixels and applies ImageNet channel normalization. The dataset remains stored locally and is not included in the project repository.

3. Methodology

Small CNN: Four convolutional stages use ReLU activations and max-pooling, followed by adaptive average pooling, dropout, and a two-class output layer. Separate runs were recorded at 5, 10, and 20 epochs.

VGG-16 transfer learning: ImageNet-pretrained VGG-16 convolutional features are frozen and extracted once for the dataset. The cached features are used to train a classifier with a 256-unit hidden layer, ReLU, dropout, and a two-class output layer. This is feature extraction rather than end-to-end fine-tuning.

4. Recorded Validation Results

Model	Validation accuracy
Small CNN — 5 epochs	73.02%
Small CNN — 10 epochs	80.84%
Small CNN — 20 epochs	87.80%
VGG-16 feature extraction	98.80%

The recorded results come from the existing training runs. The validation splits should be checked for identity before interpreting the comparison as a controlled head-to-head experiment.

5. Evaluation and Visual Results

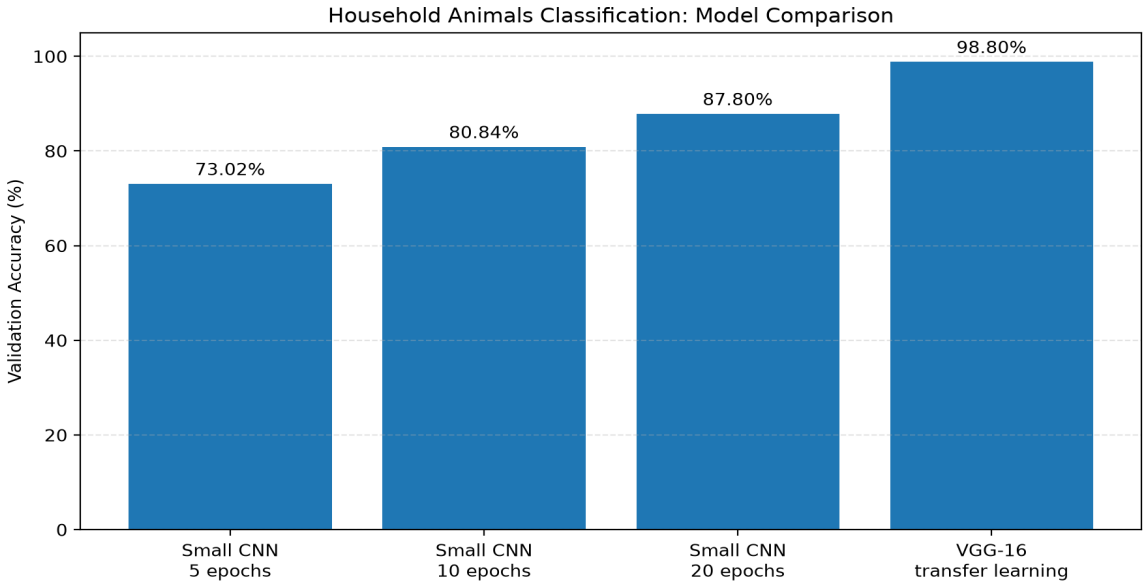


Figure 1. Recorded validation accuracy across the four experiments.

Confusion Matrices

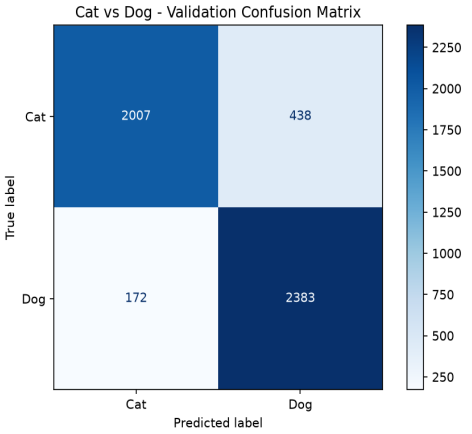


Figure 2. Small CNN confusion matrix (20-epoch model).

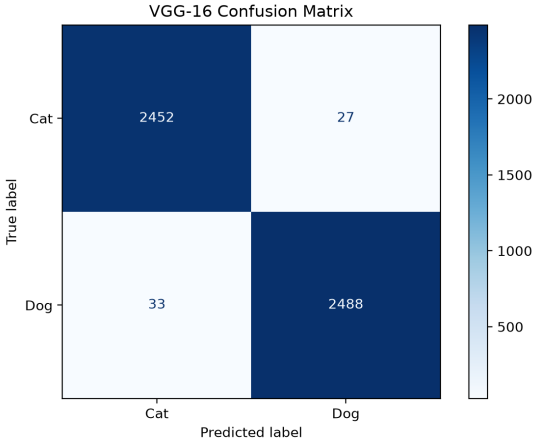


Figure 3. VGG-16 confusion matrix.

6. Discussion and Limitations

The custom CNN's recorded validation accuracy increased as the training duration increased. The VGG-16 feature-extraction experiment recorded the highest validation accuracy. Pretrained features may provide useful representations for distinguishing visual patterns in animal images. However, the models use different image resolutions and preprocessing, and their validation splits may differ. The VGG-16 training history also showed very high training accuracy and increasing validation loss, so overfitting should be considered. Validation accuracy alone does not establish performance on new datasets or guarantee reliable predictions for

every image.

7. Conclusion

The project demonstrates a complete image-classification workflow using a custom CNN and VGG-16 transfer learning, including training, cached feature extraction, evaluation, confusion matrices, and image prediction. The recorded VGG-16 result is promising. Future work should compare both models on an identical validation split, evaluate on an independent test set, and examine misclassified images.

Implementation Tools

Python 3.14.4, PyTorch 2.14.1+cpu, torchvision 0.29.1+cpu, Pillow 12.3.0, NumPy 2.5.2, Matplotlib 3.11.2, scikit-learn 1.9.1, and ReportLab 5.0.1.